

(Chia)
 - Lớp có thể chỉ là một phần / Lớp kế thừa interface qua: extend
 - Interface: phải
 - Lớp triển khai interface qua: implement

Class
 ↳ class extends
 class

interface
 ↳ class extends
 interface

write interface
 ↳ implement
 class

1 lớp có thể "nhúng" interface

VD: Chúng ta có interface Mail chứa các method sendMail(), log()

Class MailImpl sẽ implements Mail và triển khai các method mà Mail khai báo.

Nguyễn Thị Thanh Nhân

```
public interface Mail {
```

13/09/2017

13/09/2017

```
void sendMail();
```

```
void log(); }
```

```
public class MailImpl implements Mail {
```

@Override

```
public void sendMail() {
```

```
System.out.println("Đang gửi mail..."); }
```

@Override

```
public void log() {
```

```
System.out.println("Đang ghi log..."); }
```

```
}
```



TÂN TIẾN


```
public class Main {
```

```
    public static void main (String[] args) {
```

```
        Mail mail = new MailImpl();
```

```
        mail.sendMail();
```

```
        mail.log();
```

```
    }
```

bird:

```
public interface Shape {
```

```
    double getArea();
```

```
    double getPerimeter();
```

```
    String color = "red";
```

```
    void draw();
```

```
public class Rectangle implements Shape {
```

```
    private double width;
```

```
    private double height;
```

```
    public Rectangle (double width, double height) {
```

```
        this.width = width;
```

```
        this.height = height;
```

```
    @Override
```

```
    public double getArea() {
```

```
        return width * height;
```


@Override

```
public double getPerimeter () {  
    return 2 * (width + height);  
}
```

@Override

```
public void draw () {  
    System.out.println ("Draw" + color + " rectangle");  
}
```

```
public class Circle implements Shape {  
    private double radius;  
    public Circle (double radius) {  
        this.radius = radius;  
    }  
}
```

```
public class ShapeApp {  
    public static
```

@Override

```
public double getArea () {  
    return Math.PI * radius * radius;  
}
```

@Override

```
public double getPerimeter () {  
    return 2 * Math.PI * radius;  
}
```

```
public class Main {  
    public static void main (String [] args) {  
        ShapeApp shapeApp = new ShapeApp (10);  
        shapeApp.draw ();  
        System.out.println ("Area: " + shapeApp.getArea ());  
        System.out.println ("Perimeter: " + shapeApp.getPerimeter ());  
    }  
}
```



```

Shape rect = new Rectangle();
rect.draw();
System.out.println("----");
Shape circle = new Circle();
circle.draw();
}
}

```

Bài 3: abstract class PhươngTiênDiChuyen {

```

    protected String loaiPhuongTien;
    protected HangSanXuat hangSanXuat;

    public PhươngTiênDiChuyen (String loaiPhuongTien, HangSanXuat hangSanXuat) {
        this.loaiPhuongTien = loaiPhuongTien;
        this.hangSanXuat = hangSanXuat;
    }

    public String layTenHangSanXuat() {
        return hangSanXuat.getTenHangSanXuat();
    }

    public void batDau() {
        System.out.println("Bắt đầu");
    }

    public void tangTa() {
        System.out.println("Tăng tốc");
    }

    public void dungLai() {
        System.out.println("Dừng lại");
    }
}

```




```
public abstract double layVanTa ();
```

```
class HạngSanKhua {
```

```
private String tenHangSanKhua;
```

```
tenQuaGua;
```

```
public
```

```
public class MayBay extends PhuongTienDiChuyen {
```

```
private String loai NhenKen;
```

```
public MayBay (Hang sr, hang sr, string loai nhenken) {
```

```
super ("May Bay", hang sr);
```

```
this.loaiNhenKen = loaiNhenken;
```

```
public void Catcanh () {
```

```
System.out.println ("May bay cat canh");
```

```
public void Hacanh () {
```

```
System.out.println ("Maybay ha canh");
```

@ Override

```
public double layVT () {
```

```
return 800.0;
```

```
}
```

```
public class Xe Oto extends PhuongTienDiChuyen {
```




```

private String loaiNhanLen;
public XeOto (Hag sx, Hag sn, String loaiNhanLen) {
    super ("Xe Oto", Hag sx);
    this.loaiNhanLen = loaiNhanLen;
}

```

@Override

```

public double layVT () {
    return 100.0;
}

```

```

? public class XeDap extends PhuongTienDiChuyen {
    public XeDap (Hag sx, Hag sn) {
        super ("Xe Dap", Hag sx);
    }
}

```

@Override

```

public double layVT () {
    return 10.0;
}

```

class main {

```

    public static void main (String [] args) {
        Hag sx hl = new Hag sx ("Boeing", "USA");
        MayBay mb = new MayBay (hl, "Xag MayBay");
        System.out.println ("Hag SX: " + mb.layTenHag sx());
    }
}

```



```

System.out.println(" vận tốc : " + mb.getVanta() + " km/h");
mb.setColor();
}
}

```

Bài 4: public class Car {

private String color;

double speed;

public Car (String color, double speed) {

this.color = color;

this.speed = speed;

void accelerate() {

System.out.println(" Xe đang tăng tốc ...");

public String getColor() {

return color;

public double getSpeed() {

return speed;

}

public class BMW extends Car {

public BMW (String color, double speed) {
super (color, speed);

@Override




```
public void accelerate () {
```

```
    this.speed += 20;
```

```
    System.out.println ("BMW tag to: " + this.color + ", " + this
```

```
    speed);
```

```
} @Override
```

```
public String toString () {
```

```
    return "BMW [Mau sac: " + color + ", " + "Toc do: " + speed + "]\n";
```

```
}
```

```
public class Civic extends Car {
```

```
    public Civic (String color, double speed) {
```

```
        super (color, speed);
```

```
    } @Override
```

```
    public void accelerate () {
```

```
        this.speed += 10;
```

```
        System.out.println ("Civic tag to: " + this.color + ", " + this
```

```
        speed);
```

```
    } @Override
```

```
    public String toString () {
```

```
        return "Civic [Mau sac: " + color + ", " + "Toc do: " + speed + "]\n";
```




```
class main {
```

```
    public static void main (String[] args) {
```

```
        BMW BMW car1 = new BMW ("đen", 100);
```

```
        Civic car2 = new Civic ("Trắng", 80);
```

```
        car1.accelerate();
```

```
        System.out.println("car 1 to String()");  
        (" " " ");
```

```
        car2.accelerate();
```

```
        System.out.println("car 2 to String()");
```